



Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

Agentic AI Applications with Codex

TGS Ref No: TGS-2023041081

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 12.1

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
9.0	4 Sep 2026	Technical Codex deck aligned to TaskBoard labs.	Dr. Alfred Ang
10.0	22 Sep 2026	Three-topic concept flow.	Dr. Alfred Ang
11.0	23 Sep 2026	Four-topic practical course with 13 labs.	Dr. Alfred Ang
12.0	24 Sep 2026	Rebuilt as one startup journey (Cook & Bake Academy): Day 1 validates with Chat and Work and builds with Codex; Day 2 operates and markets with Work. 18 labs in their own folders; SQLite RAG assistant; skills.sh, custom skills, hooks; 2-hour assessment from 4:00 PM Day 2.	Dr. Alfred Ang
12.1	24 Sep 2026	Certification & TRAQOM Survey slide at the front and end; AI engineering timeline dated by when each practice took hold (context engineering 2024, harness engineering 2025); harness shown as the body / OS around the model with its components; agent loop as context, plan, action, verify, repeat; new What is a Plugin? slide.	Dr. Alfred Ang

TABLE OF CONTENTS

Course Overview	5
Learning Outcomes	5
Daily Schedule	5
Day 1 — Validate and Build.....	5
Day 2 — Operate and Grow.....	5
Topic-by-topic Breakdown	6
Topic 1 — Fundamentals: Chat, Work and Codex · Slides 17–57.....	6
Lab 1 — Validate the Idea with Chat and Work · Slides 29–35.....	7
Lab 2 — Plan and Build the Site with /plan · Slides 40–43.....	7
Lab 3 — Project Rules and a Sign-up Form for Every Course · Slides 47–50.....	7
Lab 4 — Publish the Site: GitHub Pages and Sites · Slides 52–56.....	7
Topic 2 — Tools and the SQLite RAG Assistant · Slides 58–87.....	7
Lab 5 — Turn the Brochures into a SQLite Knowledge Base · Slides 65–68.....	8
Lab 6 — Build the Course Assistant and Drive It with /goal · Slides 70–74.....	8
Lab 7 — Add ChatGPT Mode, Then Try to Break It · Slides 78–81.....	8
Lab 8 — QA the Whole Site with @Computer Use · Slides 83–86.....	8
Topic 3 — Skills and Hooks · Slides 88–117.....	8
Lab 9 — Install Community Skills from skills.sh · Slides 94–99.....	8
Lab 10 — Create Custom Codex Skills · Slides 102–107.....	9
Lab 11 — Guard Rails That Run Themselves · Slides 111–116.....	9
Topic 4 — Running the Academy with ChatGPT Work · Slides 118–143.....	9
Lab 12 — Price the Courses and Find Break-even · Slides 123–126.....	9
Lab 13 — Put the Term on the Calendar · Slides 127–130.....	9
Lab 14 — Enrol the Learners and Automate Reminders · Slides 132–136.....	10
Lab 15 — Customer Service with a ChatGPT Skill · Slides 138–142.....	10
Topic 5 — Digital Marketing · Slides 144–175.....	10
Lab 16 — Marketing Know-how as Skills, Then the Flyer · Slides 148–152.....	10
Lab 17 — A Multi-Agent Content Studio · Slides 156–160.....	10
Lab 18 — Lead Magnet, Newsletter and the Open-House Campaign · Slides 162–166.....	11
Resources Required	11
Assessment	11

Course Overview

A two-day, hands-on WSQ course in which learners take one business — Cook & Bake Academy Singapore, a cooking and baking school — from an idea to a fully operating company using the three agentic surfaces in the ChatGPT desktop app: ChatGPT for quick thinking, ChatGPT Work for research and operations, and Codex for building software. Day 1 validates the market and builds the website, sign-up forms and a SQLite RAG course assistant with Codex. Day 2 runs and markets the business with ChatGPT Work and a team of Codex agents.

Course code TGS-2023041081 · 2 days · 16 hours including a 2-hour assessment · 175-slide deck v12.1 · 18 labs, each in its own folder.

Learning Outcomes

By the end of this course, participants will be able to:

- **LO1.** Analyse agentic AI applications — ChatGPT, ChatGPT Work and Codex — and their strengths, limitations and suitability for a business problem.
- **LO2.** Correlate the design of an agent's instructions, tools and retrieval with the efficiency and quality of the result.
- **LO3.** Assess the effectiveness, safety and reliability of agent-built work using golden-question evaluation, hooks, Computer Use testing and human review.
- **LO4.** Evaluate comparative effectiveness and recommend a governed way to run agentic AI applications in a real business, using observable evidence.

Daily Schedule

Both days run 9:00 AM – 6:00 PM with two 10-minute tea breaks and a 45-minute lunch. The Slides column gives the deck pages for each teaching block.

Day 1 — Validate and Build

Time	Duration	Topic / Activity	Method	Slides
9:00 AM–9:20 AM	20 min	Welcome, attendance, trainer, ground rules, learning outcomes, the Cook & Bake scenario, lesson plan, assessment briefing	Presentation, discussion	1–16
9:20 AM–9:50 AM	30 min	Topic 1 — evolution of AI engineering, harness engineering and the agent loop, Chat, Work and Codex, GPT-6 models, desktop app, plugins, Work research, @Sites	Presentation, live demo	17–28
9:50 AM–10:20 AM	30 min	Lab 1 — Validate the idea with Chat and Work	Hands-on lab	29–35
10:20 AM–10:30 AM	10 min	Tea break	—	—
10:30 AM–10:40 AM	10 min	Codex: projects, folders and permissions, the 7-step workflow, /plan	Presentation, live demo	36–39
10:40 AM–11:15	35 min	Lab 2 — Plan and build the site with /plan	Hands-on lab	40–43

AM				
11:15 AM–11:20 AM	5 min	Context engineering with AGENTS.md; a sign-up form with no backend	Presentation	44–46
11:20 AM–11:50 AM	30 min	Lab 3 — Project rules and a sign-up form for every course	Hands-on lab	47–50
11:50 AM–11:55 AM	5 min	Publishing: GitHub Pages or Sites	Presentation	51
11:55 AM–12:20 PM	25 min	Lab 4 — Publish the site: GitHub Pages and Sites	Hands-on lab	52–56
12:20 PM–12:25 PM	5 min	Topic 1 recap	Discussion	57
12:25 PM–1:10 PM	45 min	Lunch	—	—
1:10 PM–1:25 PM	15 min	Topic 2 — slash commands, /goal, RAG, SQLite in the browser, architecture	Presentation, live demo	58–64
1:25 PM–1:50 PM	25 min	Lab 5 — Turn the brochures into a SQLite knowledge base	Hands-on lab	65–68
1:50 PM–1:55 PM	5 min	Measuring before trusting: the golden-question set	Presentation	69
1:55 PM–2:30 PM	35 min	Lab 6 — Build the course assistant and drive it with /goal	Hands-on lab	70–74
2:30 PM–2:35 PM	5 min	Search mode vs ChatGPT mode; keys and static sites	Presentation	75–77
2:35 PM–3:00 PM	25 min	Lab 7 — Add ChatGPT mode, then try to break it	Hands-on lab	78–81
3:00 PM–3:05 PM	5 min	The Computer Use tool	Presentation	82
3:05 PM–3:30 PM	25 min	Lab 8 — QA the whole site with @Computer Use	Hands-on lab	83–86
3:30 PM–3:40 PM	10 min	Tea break	—	—
3:40 PM–3:45 PM	5 min	Topic 2 recap	Discussion	87
3:45 PM–3:55 PM	10 min	Topic 3 — agent skills, where skills live, SKILL.md vs AGENTS.md, skills.sh	Presentation, live demo	88–93
3:55 PM–4:20 PM	25 min	Lab 9 — Install community skills from skills.sh	Hands-on lab	94–99
4:20 PM–4:25 PM	5 min	Creating your own skill; anatomy of kb-update	Presentation	100–101
4:25 PM–4:50 PM	25 min	Lab 10 — Create custom Codex skills	Hands-on lab	102–107
4:50 PM–5:00 PM	10 min	Hooks, deploy gates and scheduling by asking	Presentation	108–110
5:00 PM–5:35 PM	35 min	Lab 11 — Guard rails that run themselves	Hands-on lab	111–116
5:35 PM–6:00 PM	25 min	Day 1 review, Q&A and catch-up	Discussion	117

Day 2 — Operate and Grow

Time	Duration	Topic / Activity	Method	Slides
9:00 AM–9:20 AM	20 min	Topic 4 — running the academy, what	Presentation,	118–122

		Work does, connectors, write-access safety	live demo	
9:20 AM–9:55 AM	35 min	Lab 12 — Price the courses and find break-even	Hands-on lab	123–126
9:55 AM–10:25 AM	30 min	Lab 13 — Put the term on the calendar	Hands-on lab	127–130
10:25 AM–10:35 AM	10 min	Tea break	—	—
10:35 AM–10:40 AM	5 min	Scheduled and event-triggered tasks	Presentation	131
10:40 AM–11:15 AM	35 min	Lab 14 — Enrol the learners and automate reminders	Hands-on lab	132–136
11:15 AM–11:20 AM	5 min	Skills in ChatGPT	Presentation	137
11:20 AM–11:55 AM	35 min	Lab 15 — Customer service with a ChatGPT skill	Hands-on lab	138–142
11:55 AM–12:00 PM	5 min	Governance: what you just automated	Discussion	143
12:00 PM–12:45 PM	45 min	Lunch	—	—
12:45 PM–1:00 PM	15 min	Topic 5 — the funnel for a course business, marketing skills and image generation	Presentation	144–147
1:00 PM–1:30 PM	30 min	Lab 16 — Brand kit, flyer and social cards	Hands-on lab	148–152
1:30 PM–1:45 PM	15 min	Custom agents in Codex, the content studio, making the YouTube video	Presentation, live demo	153–155
1:45 PM–2:30 PM	45 min	Lab 17 — A multi-agent content studio	Hands-on lab	156–160
2:30 PM–2:35 PM	5 min	Leads, newsletters and consent	Presentation	161
2:35 PM–3:15 PM	40 min	Lab 18 — Lead magnet, newsletter and the open-house campaign	Hands-on lab	162–166
3:15 PM–3:25 PM	10 min	Tea break	—	—
3:25 PM–4:00 PM	35 min	Course summary: why delegate to agents, how each feature is triggered, idea to operational; Q&A and assessment briefing	Discussion	167–169
4:00 PM–5:00 PM	60 min	Written Assessment (WA) — open book, 1 hour	Assessment	—
5:00 PM–6:00 PM	60 min	Practical Performance (PP) — 1 hour; TRAQOM, attendance, submit, sign the record	Assessment	170–175

Topic-by-topic Breakdown

Topic 1 — Fundamentals: Chat, Work and Codex • Slides 17–57

- How AI engineering got here, and why the harness matters

- OpenAI's products, the GPT-6 models and the desktop app
- Validate the idea: Chat, Work research and Sites
- Build the site with Codex, add sign-ups, publish it

Lab 1 — Validate the Idea with Chat and Work • Slides 29–35

- **Surface:** Chat → ChatGPT Work → @Sites
- **Time:** 30 minutes (Day 1)
- **Features practised:** Chat brainstorm · Work research with plan review · @Sites HTML report
- **Output:** A private HTML market report on Sites, and market-brief.md for Codex
- **Evidence:** Chat returned five research questions you used in Prompt B. market-brief.md is saved in your cook-and-bake folder.
- **Lab folder:** labs/lab-01-validate-the-idea/

Lab 2 — Plan and Build the Site with /plan • Slides 40–43

- **Surface:** Codex
- **Time:** 35 minutes (Day 1)
- **Features practised:** Add a folder · GPT-6 Sol · /plan · the 7-step workflow
- **Output:** A running site: hero with photos, 20 course cards, filters, search, campuses
- **Evidence:** Plan mode showed numbered steps AND questions before any edit. No horizontal scroll at 375px wide.
- **Lab folder:** labs/lab-02-plan-and-build-the-site/

Lab 3 — Project Rules and a Sign-up Form for Every Course • Slides 47–50

- **Surface:** Codex
- **Time:** 30 minutes (Day 1)
- **Features practised:** /init · AGENTS.md · static-site forms · consent
- **Output:** AGENTS.md, a validated sign-up form on every card, and a staff CSV export
- **Evidence:** AGENTS.md has the four sections and is under 60 lines. admin.html exports a CSV with the 12 columns in the spec.
- **Lab folder:** labs/lab-03-rules-and-signup-forms/

Lab 4 — Publish the Site: GitHub Pages and Sites • Slides 52–56

- **Surface:** Codex → GitHub Pages → @Sites
- **Time:** 25 minutes (Day 1)
- **Features practised:** git · GitHub Actions · GitHub Pages · @Sites access control
- **Output:** A public GitHub Pages site for customers and a private Sites copy for investors
- **Evidence:** Codex listed the files and none is a secret or an export. You can say which URL is for customers and which for investors.
- **Lab folder:** labs/lab-04-publish-the-site/

Topic 2 — Tools and the SQLite RAG Assistant • Slides 58–87

- Codex commands, /goal and tools
- RAG: answering from your own documents
- A SQLite knowledge base that runs in the browser
- ChatGPT mode, red-teaming and Computer Use QA

Lab 5 — Turn the Brochures into a SQLite Knowledge Base • Slides 65–68

- **Surface:** Codex
- **Time:** 25 minutes (Day 1)
- **Features practised:** SQLite FTS5 · official SQLite WASM · nested AGENTS.md
- **Output:** data/academy.db — 140 searchable chunks and a courses table
- **Evidence:** data/academy.db exists (about 150 KB). Running the build twice gives the same counts.
- **Lab folder:** labs/lab-05-sqlite-knowledge-base/

Lab 6 — Build the Course Assistant and Drive It with /goal • Slides 70–74

- **Surface:** Codex
- **Time:** 35 minutes (Day 1)
- **Features practised:** In-browser RAG · golden-question evaluation · /goal
- **Output:** A working course assistant that passes 30/30 golden questions
- **Evidence:** The assistant runs in the browser with no server code. "What is the cheapest course?" answers CUL-210, S\$160.
- **Lab folder:** labs/lab-06-course-assistant-and-goal/

Lab 7 — Add ChatGPT Mode, Then Try to Break It • Slides 78–81

- **Surface:** Codex
- **Time:** 25 minutes (Day 1)
- **Features practised:** OpenAI Responses API · gpt-6-luna · grounded prompt · prompt injection
- **Output:** A two-mode assistant that stays grounded under attack
- **Evidence:** git grep "sk-" finds no key anywhere in the repo. A bad key falls back to search mode with a message.
- **Lab folder:** labs/lab-07-chatgpt-mode-and-red-team/

Lab 8 — QA the Whole Site with @Computer Use • Slides 83–86

- **Surface:** Codex + Computer Use
- **Time:** 25 minutes (Day 1)
- **Features practised:** @Computer Use · evidence-based QA
- **Output:** A ranked defect table with screenshots, and one verified fix
- **Evidence:** Computer Use drove a real browser through the whole script. The top defect is fixed and the retest confirms it.
- **Lab folder:** labs/lab-08-qa-with-computer-use/

Topic 3 — Skills and Hooks • Slides 88–117

- Skills: packaged expertise, from skills.sh or your own
- Custom Codex skills with \$skill-creator
- Hooks that enforce rules and re-test automatically
- A deploy gate and a scheduled health check

Lab 9 — Install Community Skills from skills.sh • Slides 94–99

- **Surface:** Codex
- **Time:** 25 minutes (Day 1)

- **Features practised:** npx skills add · frontend-design · cybersecurity-analyst · anthropic-cybersecurity-skills
- **Output:** A polished UI, a ranked security review, and skills-lock.json
- **Evidence:** Three skills are in .agents/skills/ and skills-lock.json. You fixed the top finding (e.g. a Content-Security-Policy).
- **Lab folder:** labs/lab-09-skills-from-skills-sh/

Lab 10 — Create Custom Codex Skills · Slides 102–107

- **Surface:** Codex
- **Time:** 25 minutes (Day 1)
- **Features practised:** \$skill-installer · \$skill-creator · implicit triggering
- **Output:** \$kb-update and \$course-brochure skills, and a new course live
- **Evidence:** project-setup is installed and listed, but was not run. \$kb-update fired on its own in a new chat.
- **Lab folder:** labs/lab-10-custom-codex-skills/

Lab 11 — Guard Rails That Run Themselves · Slides 111–116

- **Surface:** Codex → GitHub Actions → Scheduled
- **Time:** 35 minutes (Day 1)
- **Features practised:** hooks.json · PreToolUse · PostToolUse · deploy gate · scheduled task
- **Output:** Two trusted hooks, a gated deploy and a Monday health check
- **Evidence:** Both hooks are trusted, and test_hooks.py passes all 9 cases. The weekly task is in Scheduled, read-only, and reports on success.
- **Lab folder:** labs/lab-11-guard-rails-that-run-themselves/

Topic 4 — Running the Academy with ChatGPT Work · Slides 118–143

- From website to operating business
- Plan mode and connectors: Drive, Calendar, Gmail
- Scheduled and event-triggered automation
- Skills in ChatGPT, and governing what you automate

Lab 12 — Price the Courses and Find Break-even · Slides 123–126

- **Surface:** ChatGPT Work + @Drive → Chat
- **Time:** 35 minutes (Day 2)
- **Features practised:** Plan mode in Work · spreadsheet artifact · @Drive
- **Output:** A pricing workbook with live formulas and a fee recommendation
- **Evidence:** Work showed a plan and you changed one step. You recomputed one course's cost per seat by hand.
- **Lab folder:** labs/lab-12-price-the-courses/

Lab 13 — Put the Term on the Calendar · Slides 127–130

- **Surface:** ChatGPT Work + @Drive + @Calendar
- **Time:** 30 minutes (Day 2)
- **Features practised:** @Calendar write access · clash detection · approve before write
- **Output:** A dedicated term calendar with every session, and a clash report

- **Evidence:** Work listed every event and waited for approval. You opened Google Calendar and saw the events.
- **Lab folder:** labs/lab-13-term-on-the-calendar/

Lab 14 — Enrol the Learners and Automate Reminders • Slides 132–136

- **Surface:** ChatGPT Work + @Drive + @Gmail → Scheduled
- **Time:** 35 minutes (Day 2)
- **Features practised:** @Gmail drafts · data checks · scheduled tasks
- **Output:** One draft per learner, a flag list, and a weekly reminder task
- **Evidence:** Only drafts were created — nothing was sent unprompted. Your one test email arrived; the weekly task is in Scheduled.
- **Lab folder:** labs/lab-14-enrol-and-remind/

Lab 15 — Customer Service with a ChatGPT Skill • Slides 138–142

- **Surface:** Chat → ChatGPT Work (@skill-creator, @Gmail, event trigger)
- **Time:** 35 minutes (Day 2)
- **Features practised:** Skills in ChatGPT · event-triggered tasks · feedback analysis
- **Output:** An enquiry-reply skill, an event-triggered drafting task and a feedback report
- **Evidence:** The skill exists and runs when you mention an enquiry. The report names BAK-105 and the warm-room complaint.
- **Lab folder:** labs/lab-15-customer-service/

Topic 5 — Digital Marketing • Slides 144–175

- A funnel for a course business
- Marketing know-how as custom skills; the flyer with \$imagegen
- A multi-agent content studio in Codex
- Lead magnet, newsletter and measuring what worked

Lab 16 — Marketing Know-how as Skills, Then the Flyer • Slides 148–152

- **Surface:** Chat → Codex (\$skill-creator, \$imagegen)
- **Time:** 30 minutes (Day 2)
- **Features practised:** custom marketing skills · \$imagegen · proofreading
- **Output:** Four marketing skills (brand-voice, flyer-design, social-post, youtube-package) and an A5 flyer, 1:1 post and 9:16 story
- **Evidence:** Four skills exist in .agents/skills/, each with a "Use when" description. In a new chat, "write an Instagram post about macarons" triggers social-post on its own.
- **Lab folder:** labs/lab-16-marketing-skills-and-flyer/

Lab 17 — A Multi-Agent Content Studio • Slides 156–160

- **Surface:** Codex (custom agents + custom skills)
- **Time:** 45 minutes (Day 2)
- **Features practised:** .codex/agents/*.toml · parallel subagents · the Lab 16 skills · reviewer agent
- **Output:** content/week-01/: 7 posts, images, a YouTube video package and a review log
- **Evidence:** Five agent threads ran; you opened at least two. Nothing was published.

- **Lab folder:** labs/lab-17-multi-agent-content-studio/

Lab 18 — Lead Magnet, Newsletter and the Open-House Campaign • Slides 162–166

- **Surface:** Codex (skills.sh) → ChatGPT Work (@Gmail, analysis)
- **Time:** 40 minutes (Day 2)
- **Features practised:** lead-magnets · seo-audit · newsletter-generation · consent · UTM
- **Output:** A lead magnet, an SEO-fixed site, a newsletter to consented contacts only, and a results report
- **Evidence:** Three marketing skills are installed and recorded in the lock file. The results report gives conversion per channel and a budget split.
- **Lab folder:** labs/lab-18-open-house-campaign/

Resources Required

- Trainer: the v12 slide deck, this Lesson Plan, the Learner Guide, the lab pack (labs/, 18 folders) and the reference app for Labs 2–11.
- Each learner: a laptop with the ChatGPT desktop app on a paid plan that includes Codex, ChatGPT Work and Sites; Node.js 22; Python 3; Git; a GitHub account; and a personal Google account for Drive, Calendar and Gmail.
- A spend-limited OpenAI API key issued by the trainer for Lab 6, revoked after class.
- Internet access to chatgpt.com, api.openai.com, github.com, skills.sh, npmjs.com, images.unsplash.com and canva.com.
- All fixtures are synthetic. Every learner email address resolves to the learner's own Gmail via plus-addressing.

Assessment

- Day 2, 4:00 – 6:00 PM: Written Assessment (WA, short-answer, 1 hour) from 4:00 PM, then Practical Performance (PP, 1 hour) from 5:00 PM.
- Open book: slides, Learner Guide and approved materials only.
- Funding criteria: minimum 75% attendance, and assessed as Competent in both the WA and the PP.
- Assessment flow: TRAQOM survey → assessment digital attendance → WA and PP → submit answers on the LMS → sign the Assessment Summary Record.
- Courseware and assessment are on the LMS: <https://lms-tms.tertiaryinfotech.com/>